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High Temperature–Low Stress Creep of Al and Al + 0.5% Fe

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*Summary***

The kinetics and structure dependence of high temperature–low stress Newtonian creep of Al and Al + 0.5% Fe have been studied and compared with the predictions of the Nabarro–Herring (N–H) theory of directional diffusion. It is shown that although the stress and temperature dependence of the Newtonian creep rate of pure Al are in agreement with the N–H theory, the observed absolute creep rates in both single crystals and polycrystals are 1500 times faster than the theory predicts. Further it is shown that Newtonian creep in Al is structure sensitive. The dislocation density increases and subgrains form during creep. The subgrain size varies inversely with stress and extrapolates to subgrain size measurements for Al at lower temperatures and higher stresses while the dislocation density is essentially stress independent. Pre-straining at low temperatures results in inverse transient creep behavior similar to that observed in cold worked metals tested at lower temperatures and higher stresses. Also, the presence of precipitates of $FeAl_3^*$ reduces the creep rate of Al by about a factor of 100 at the same stress and temperature, in spite of the fact that the grain size in the Al–5% Fe alloy is smaller by a factor of 30. These experimental facts lead us to conclude that Newtonian creep in Al deformation is controlled by a dislocation process rather than by directional diffusion. A simple theory based on the climb controlled generation of dislocations from a fixed density of sources is developed to explain the observed behavior.

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** For French and German translations of the Summary, see p. 42.

INTRODUCTION

One of the most successful theories of plastic deformation of metals and other crystalline materials pertains to high temperature–low stress creep. Just over 20 years ago Nabarro[1] and Herring[2] independently suggested that plastic flow could take place by the directional diffusion of vacancies between interfaces. Their theory predicted a Newtonian viscous creep rate (linear stress dependence) with a temperature dependence equal to that for self-diffusion. Subsequent experimental studies have shown the observed creep kinetics to be in good agreement with the theory[3]. In addition, microstructural observations[4,5] have provided direct evidence that viscous creep can occur by a directional diffusion mechanism rather than by a process such as dislocation motion. Since the first observations of Newtonian viscous creep, there have been a number of experimental studies which have used this phenomenon to determine the surface energy of metals by the zero creep method[6].

In addition to the original Nabarro–Herring (N–H) theory of diffusional creep, numerous modifications have been proposed over the last few years. These modifications include allowance for grain boundary diffusion when the grain size is small[7], consideration of transient creep in terms of dislocations acting as vacancy sources and sinks[8], inclusion of grain boundary sliding as a deformation mode contributing to the strain[9] and treatment of dislocations as the primary vacancy sources and sinks instead of grain boundaries and surfaces[10,11] These additional modifications broaden the scope of the Nabarro–Herring theory to the extent that the great bulk of the experimental data are in good agreement with the orignal theory or can be understood with one of the proposed modifications.

Perhaps the most significant deviation of theory and experiment is represented by the work of

Harper and Dorn[12,13] on the low stress–high temperature creep of Al. These authors observed a number of features which suggested that Newtonian viscous creep in Al is not controlled by a simple directional diffusion mechanism but rather is controlled by dislocation motion. These observations include: (a) identical creep rates for single crystals and polycrystals, (b) recovery of creep strain following unloading, (c) poor absolute rate agreement between the N–H theory and experiment and (d) uniformity of deformation within grains and across grain boundaries. These observations suggest that creep of Al at high temperatures and low stresses take place by dislocation climb as is generally conceded to be the case at higher stresses.

Friedel[10] suggested that the poor agreement between theory and experiment for Al is due to diffusion between dislocations, but if this is the case then the physical process occurring during creep is dislocation climb with vacancy generation and annihilation taking place at dislocations to assure a steady state. If this is indeed the case, one might expect to see some dislocation substructure generated as is found in lower temperature–higher stress creep[14]. Also the addition of any structural feature which might impede dislocation motion such as a fine dispersion of second phase should produce a change in the observed creep rate.

The above speculation on the creep process operative in pure Al during Newtonian viscous deformation provided the rationale for the present study. An attempt was made to determine the rate-controlling creep process operative in Al when deformed at high temperatures and low stresses. Etch pit observations were combined with creep kinetics measurements to determine the extent of dislocation motion during deformation. Pure Al polycrystals and single crystals were studied along with polycrystals of a two-phase Al–0.5% Fe alloy in order to determine the importance of microstructure to the creep process.

EXPERIMENTAL PROCEDURE

The chemical analyses of the material used in this investigation are shown in Table 1. Samples were machined into shape using a lubricant of 25 g of paraffin dissolved in 500 cm^3 of kerosene. The gage section of the samples was in the form of a right circular cylinder of length 3.75 cm and diameter 1.0 cm (for Al) or 0.5 cm (for Al–0.5% Fe). Single crystal specimens of Al were prepared from slabs grown by directional solidification. The gage section of the single crystal specimens was 3.75 cm long, 1.25 cm wide and approximately 0.3 cm thick. All specimens were electropolished to remove any machining damage prior to testing.

TABLE 1

Spectrochemical analyses of Al and Al–0.5% Fe (compositions reported in weight percent)

Al		*Al–0.5% Fe*	
Fe	< 0.002	Fe	0.5
Mg	< 0.001	Mg	<0.001
Cu	0.003	Si	<0.003
Al	Balance	Cu	0.001
		Mn	0.001
		Al	Balance

All creep tests were performed in uniaxial tension under constant load. Because of the small strains involved this is equivalent to constant stress. The testing atmosphere was air and temperature measurement was made using clad chromel–alumel thermocouples. The temperature control was generally ± 1 deg C during any test. Strain was measured by means of a linear variable differential transformer and a millivolt potentiometer. Deflections as small as 10^{-4} cm could be measured with a reproducibility of better than $\pm 2\%$.

Samples were annealed in the creep machine prior to testing. The annealing treatment was 36 hours at $\sim 650°$ C (0.99 T_m) under the weight of the lower specimen grip. This amounted to a stress of 0.4 p.s.i. (2.75×10^3 N/m^2) for the Al polycrystal and slightly higher stresses for the single crystals and Al–0.5% Fe.

Following annealing the grain size of the Al polycrystals was ~ 1 cm whereas the Al–0.5% Fe samples had a grain size of 0.3 mm. Room temperature optical microscopy showed the Al_3Fe precipitate in the Al–0.5 Fe alloy to be cylindrical in shape with a length of 7 μm and a diameter of ~ 2 μm. The average spacing between Al_3Fe particles was computated to be ~ 10 μm.

Structural observations were made using X-ray diffraction and optical microscopy. Samples examined in optical microscopy were etch pitted using an etchant of 50% HCl, 47% HNO_3 and 3% HF. All reported dislocation densities were determined from etch pit densities using the relation[15]

$$\rho = \frac{2N}{A} \tag{1}$$

where ρ is the dislocation density and N is the number of etch pits in area A.

EXPERIMENTAL RESULTS AND DISCUSSION

The steady state creep data for pure Al are listed in Table 2 and are shown in Fig. 1. The results clearly show Newtonian viscous creep behavior at stresses below about 10 p.s.i. (6.89×10^4 N/m^2). The creep rates at stresses above 10 p.s.i. (6.89×10^4 N/m^2), where power law creep is observed, are in good agreement with extrapolated data of other investigators[16,17]. The data include strain rates for both single crystals and polycrystalline samples. Three types of tests were performed on polycrystalline samples. Generally samples were annealed and creep tested at a single stress. However, several samples were annealed, tested at one level of stress into steady state, the stress changed and the test continued until steady state was again achieved. These tests are referred to as cyclic tests. Observed steady state creep rates do not depend on the prior testing history of the sample. The third group of tests were iso-stress creep tests performed on samples which had been annealed, pre-strained approxi-

TABLE 2

Steady state creep data for pure Al samples

Applied stress (*p.s.i.*)	(*N/m*^{2})	*Temperature* (*K*)	*Steady state creep rate* (*min*$^{-1}$)	(*s*$^{-1}$)
2.1	1.45×10^4	923	2.5×10^{-7}	4.16×10^{-9}
3.2*	2.20×10^4	918	3.7×10^{-7}	6.16×10^{-9}
5.0	3.44×10^4	925	6.4×10^{-7}	1.07×10^{-8}
5.1†	3.52×10^4	923	4.7×10^{-7}	7.85×10^{-9}
6.2†	4.27×10^4	918	7.7×10^{-7}	1.28×10^{-8}
7.0	4.82×10^4	923	9.9×10^{-7}	1.65×10^{-8}
[8.1†	5.58×10^4	922	3.1×10^{-6}	5.16×10^{-8}
17.3	11.9×10^4	922	6.9×10^{-6}	1.15×10^{-7}
[8.5	5.85×10^4	925	2.4×10^{-6}	4.00×10^{-8}
18.1	12.5×10^4	925	8.4×10^{-6}	1.40×10^{-7}
8.3	5.72×10^4	925	1.0×10^{-6}	1.67×10^{-8}
8.5	5.85×10^4	923	2.0×10^{-6}	3.33×10^{-8}
8.7*	6.00×10^4	914	1.4×10^{-6}	2.33×10^{-8}
[9.0	6.20×10^4	920	1.3×10^{-6}	2.16×10^{-8}
18.9	13.0×10^4	920	4.8×10^{-6}	8.00×10^{-8}
9.1	6.27×10^4	921	1.4×10^{-6}	2.33×10^{-8}
9.6	6.60×10^4	916	2.4×10^{-6}	4.00×10^{-8}
[9.8	6.75×10^4	918	2.1×10^{-6}	3.50×10^{-8}
18.1	12.5×10^4	918	6.8×10^{-6}	1.13×10^{-7}
9.8	6.75×10^4	918	1.7×10^{-6}	2.83×10^{-8}
11.7	8.07×10^4	920	1.7×10^{-6}	2.83×10^{-8}
14.2	9.80×10^4	925	3.5×10^{-6}	5.84×10^{-8}
17.1	11.8×10^4	923	4.6×10^{-6}	7.66×10^{-8}
20.3	14.0×10^4	923	9.4×10^{-6}	1.57×10^{-7}
20.3	14.0×10^4	921	4.4×10^{-6}	7.35×10^{-8}
22.6	15.6×10^4	924	3.9×10^{-5}	6.50×10^{-7}

* Single crystals.
[Denotes cyclic test series.
† Prestrained 2%.

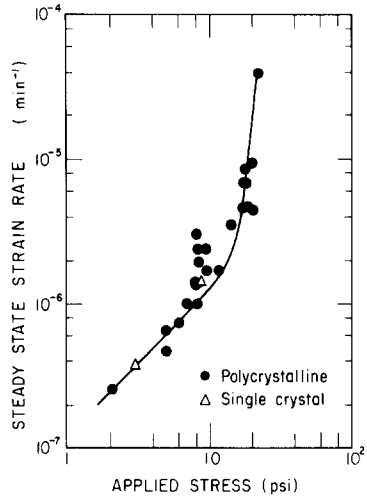

Fig. 1. Steady state strain rates of Al as a function of applied stress at temperatures between 914K and 925K.

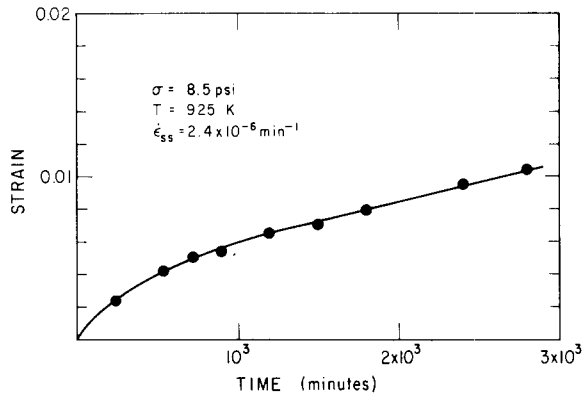

Fig. 2. Typical creep curve for Al tested at high temperature and low stress.

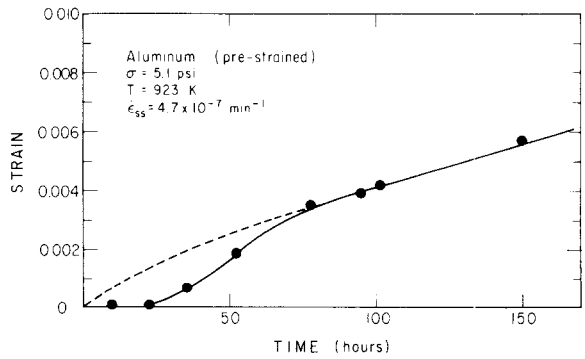

Fig. 3. Typical creep curve for Al prestrained at room temperature prior to testing at high temperature and low stress.

TABLE 3

Determination of creep activation energy for pure Al

Applied stress (*p.s.i.*)	(N/m^2)	*Temperature* (*K*)	*Creep rate* (min^{-1})	(s^{-1})	*Activation energy* (*kcal/mol*)
8.1	5.58×10^4	922	3.1×10^{-6}	5.16×10^{-8}	32.5
8.1	5.58×10^4	901	2.0×10^{-7}	3.33×10^{-8}	27.0
8.1	5.58×10^4	921	2.5×10^{-6}	4.17×10^{-8}	
17.3	1.19×10^5	921	6.9×10^{-6}	1.15×10^{-7}	37.0
17.3	1.19×10^5	904	4.7×10^{-6}	7.84×10^{-8}	37.7
17.3	1.19×10^5	922	7.1×10^{-6}	1.18×10^{-7}	
5.1	3.51×10^4	922	2.7×10^{-7}	4.50×10^{-9}	39.6
5.1	3.51×10^4	903	1.7×10^{-7}	2.83×10^{-9}	37.6
5.1	3.51×10^4	922	2.6×10^{-7}	4.33×10^{-9}	
					Average 35.2 kcal/mol

mately 2% at room temperature, re-annealed and then creep tested. The pre-strain had no influence on the steady state characteristics but, as discussed below, did affect the transient creep behavior.

The results of creep tests on single and polycrystalline specimens agree well with the data of others[12,13]. Also, as found by Harper and Dorn[12], the strain rates for single crystals agree closely with rates observed for polycrystalline samples. Comparison of the observed rates with the N–H model indicates that Al creeps some 1000–1500 times faster than the directional diffusion theory would predict.

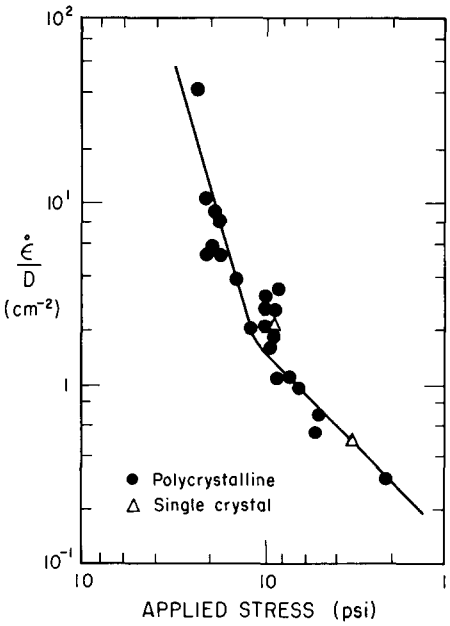

Fig. 4. Temperature-compensated steady state creep rates of Al as a function of applied stress.

The creep curve for a polycrystalline specimen shown in Fig. 2 is typical of that of both single crystal and polycrystalline samples. The primary strain in evidence in Fig. 2 was observed in all isostress creep tests. A typical creep curve for one of the pre-strained samples is shown in Fig. 3. The characteristic sigmoidal shape of the creep curve is

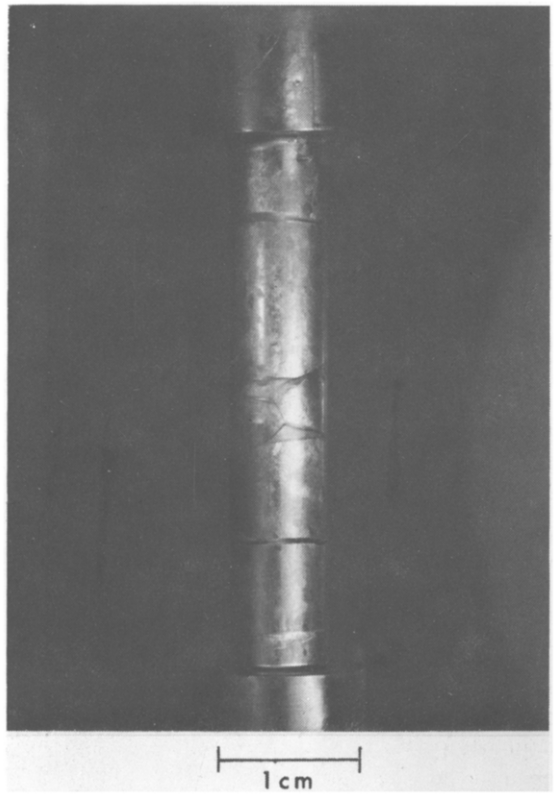

Fig. 5. Photograph of a polycrystalline Al creep sample tested at 925K at an applied stress of 5 p.s.i. (3.44×10^4 N/m^2) showing grain boundary sliding and subgrain formation.

similar to that observed for samples pre-strained prior to creep testing at lower temperatures and higher stresses[18].

The creep activation energy was determined by measuring the steady state creep rates before and after a small temperature change (~ 20 deg C). The data obtained, uncorrected for the temperature dependence of the elastic modulus, are shown in Table 3. The observed activation energy (35.2 kcal/mol) is in good agreement with the value for self diffusion in Al[19] (33.6 kcal/mol). As the temperature dependence of the creep rate is close to that for self diffusion, the data in Fig. 1 can be corrected for minor differences in testing temperature by plotting $\dot{\varepsilon}/D$ where D is the self diffusion coefficient. This temperature compensation does not significantly alter the observed Newtonian viscous behavior (see Fig. 4).

There are several significant microstructural features associated with the viscous deformation of Al. The photograph of one of the polycrystalline samples shown in Fig. 5 illustrates two important characteristics. First, considerable grain boundary sliding and grain migration are observed. Although it is clear that the sliding at grain boundaries contributes to the overall strain, the close correspondence of creep rates for single and polycrystals indicates that grain boundary sliding does not make a significant contribution to the observed steady state strain rate. This is probably a result of the fact that during primary creep the boundaries generally migrate to positions nearly normal to the applied stress where the resolved shear stress is zero. Thus the grain boundary sliding contribution to the strain rate appears limited to the primary creep regime where the boundaries are inclined to the stress axis. In support of this conclusion is the fact that polycrystalline samples exhibited more extensive primary creep deformation than single crystals when tested at equivalent stresses and temperatures.

A second microstructural feature involves the generation of a subgrain network. These subgrains are visible on the sample surface after testing and readily apparent on polished and etch pitted samples (see Fig. 6). The subgrain structure was present in both single and polycrystals. No subgrains were evident in undeformed control samples that were given only the standard annealing treatment. The variation of subgrain size with applied stress is shown in Fig. 7 along with other data for higher stresses and lower temperatures[20,21]. There is good

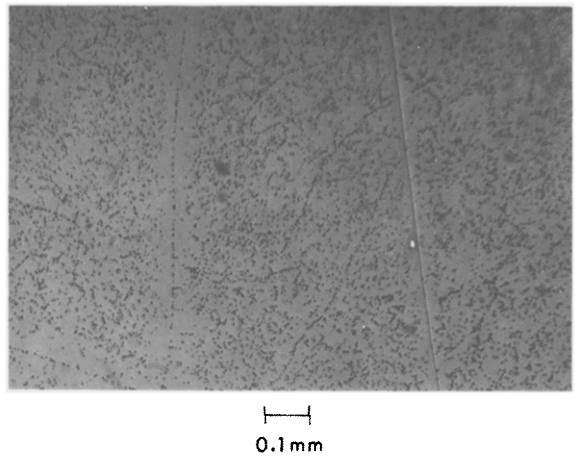

Fig. 6. Etch pit pattern of an Al sample tested at 9.6 p.s.i. (6.6×10^4 N/m^2) and 916K. The vertical boundary in the right side of the photograph is a high-angle grain boundary.

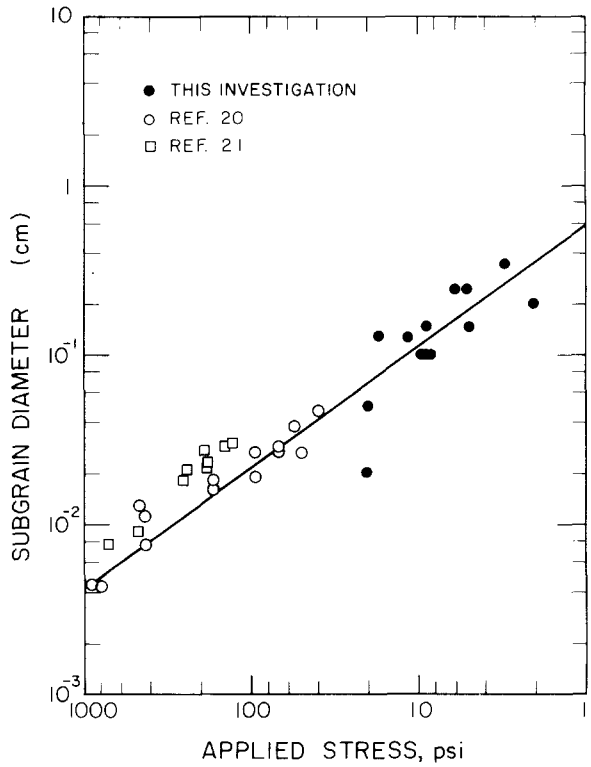

Fig. 7. Subgrain diameter as a function of applied stress for Al tested under creep conditions.

agreement between the various sets of data with the subgrain size varying inversely with applied stress. Lang topography and Laue back reflection experiments were performed to substantiate that the subgrains apparent in the etch pit pattern actually correspond to a subgrain network.

Dislocation densities as a function of applied

TABLE 4

Dislocation density as a function of applied stress for pure Al

Applied stress (*p.s.i.*)	(N/m^2)	*Dislocation density* (cm^{-2})
0 (annealed)	0	1,320
2.1	1.45×10^4	3,880
3.1*	2.12×10^4	4,880
8.3*	5.72×10^4	5,100
8.7	6.00×10^4	9,690
9.6	6.60×10^4	8.200
11.7	8.07×10^4	4,080
14.2	9.80×10^4	4,280
17.1	11.8×10^4	5.720
20.3	14.0×10^4	5,990
20.3	14.0×10^4	8,400

* Samples were cyclically stressed during testing. The stress listed is the final stress.

stress are listed in Table 4. The significant feature of these results is the apparent stress independence of the dislocation density over the viscous creep range. Although most previous investigators have observed stress dependent dislocation densities during creep deformation, Barrett[22] and Stang[23] have observed that in creep of Fe–3% Si the stress dependence of the dislocation density decreases as the temperature is increased and the applied stress decreased.

The increase in dislocation density over the as-annealed value along with the stress dependent subgrain structure indicates that dislocation motion is an important deformation mechanism operative during viscous creep of Al. Other evidence supporting this conclusion are:

(1) The steady state strain rates are some 1500 times greater than the predicted directional diffusion rates; even if one assumes that diffusion occurs between subgrain boundaries the predicted rates are still 30–40 times too slow

(2) Single crystals creep at the same rate as polycrystals

(3) Pre-straining samples results in inverse transient creep behavior.

If we assume that viscous creep in Al is actually controlled by dislocation movement, then any proposed dislocation mechanism must predict three characteristic features of the deformation process. First, the theory must predict a linear stress dependence for the creep rate along with self diffusion temperature dependence. Secondly, the theory should predict a stress independent dislocation density. Lastly, the absolute magnitude of the creep rate predicted by theory should compare favorably with the observed rate. This last point is often neglected in comparison of theory with experiment yet must be satisfied to assure a satisfactory interpretation.

To analyze the Newtonian viscous behavior of pure Al in terms of dislocation motion we first focus attention on the time rate of change of the dislocation density, $\dot{\rho}$. We assume that dislocation generation occurs by a diffusion controlled climb process which is driven by the applied stress. Taking the generation process to occur at a fixed number of sources per unit volume (independent of the applied stress), the rate of dislocation generation, $\dot{\rho}_+$, can be written

$$\dot{\rho}_+ = \rho_0 \frac{v_c}{x} \tag{2}$$

where ρ_0 is the fixed dislocation source length per unit volume, v_c is the climb velocity and x is the distance over which climb must occur to create glissile dislocations.

Taking the usual expression for the climb velocity[24]

$$v_c = \frac{Db^2\sigma}{kT} \tag{3}$$

and

$$x = \frac{Gb}{\sigma} \tag{4}$$

we obtain

$$\dot{\rho}_+ = \frac{\rho_0 Db\sigma^2}{kTG}. \tag{5}$$

Whereas the assumptions of a fixed density of dislocation sources and a diffusion controlled generation rate cannot be rigorously defended, they are perhaps not unreasonable assumptions considering the experimentally observed stress independent dislocation density. Models based on first-order dislocation generation kinetics and second-order annihilation kinetics as generally used for creep conditions[25] at lower temperatures and higher stresses are not capable of predicting the observed behavior.

Dislocation annihilation is assumed to occur only at subgrain boundaries. With this assumption the annihilation rate, $\dot{\rho}_-$, can be written

$$\dot{\rho}_- = \frac{\rho v_g}{\lambda} \tag{6}$$

where ρ is the glissile dislocation density, v_g is the

glide velocity and λ is the subgrain diameter. Following the data in Fig. 7, λ can be approximated as

$$\lambda = \lambda_0/\sigma \tag{7}$$

where λ_0 is a constant. Also, for a number of different rate controlling glide processes (jogged screw dislocations, viscous drag, etc.), a good approximation[26] for v_g in the stress and temperature range of interest is

$$v_g = v_0 \sigma \tag{8}$$

where v_0 is a temperature dependent constant. Combining the above we have

$$\dot{\rho}_- = \frac{\rho v_0 \sigma^2}{\lambda_0}. \tag{9}$$

The expression for $\dot{\rho}$ can then be written

$$\dot{\rho} = \dot{\rho}_+ - \dot{\rho}_- = \frac{\rho_0 D b \sigma^2}{kTG} - \frac{\rho v_0 \sigma^2}{\lambda_0}. \tag{10}$$

At steady state $\dot{\rho} = 0$ and

$$\rho = \frac{\rho_0 D b \lambda_0}{kTG v_0}. \tag{11}$$

Note that ρ is independent of stress consistent with the data in Table 4 and depends on temperature primarily through the ratio D/v_0. If glide is diffusion controlled then $v_0 \propto D$ and ρ is essentially temperature independent. If glide is phonon drag controlled, however, then D/v_0 will, to a first approximation, follow the temperature dependence of D. Unfortunately, no data exist on the temperature dependence of ρ in the range of interest.

Combining eqn. (11) with

$$\dot{\varepsilon}_s = \rho b v_g \tag{12}$$

we have a simple expression for the strain rate

$$\dot{\varepsilon}_s = \frac{\rho_0 D b^2 \sigma \lambda_0}{kTG}. \tag{13}$$

Note that this expression predicts the observed behavior of $\dot{\varepsilon}_s$, namely self diffusion temperature dependence and linear stress dependence. To examine the absolute rate prediction of eqn. (13), we take the following values:

$\rho_0 = 10^3$ cm^{-2} (essentially the as-annealed dislocation density)
$D = 1.3 \times 10^{-8}$ cm^2/sec (923 K)
$b = 2.8 \times 10^{-8}$ cm
$\lambda_0 = 1$ cm. p.s.i. $= 68.9$ N/m
$kT = 1.27 \times 10^{-13}$ erg (923 K)
$G = 1.6 \times 10^7$ p.s.i. $= 1.1 \times 10^{11}$ N/m^2

and predict the creep rate at 923° C and 5 p.s.i. (3.44×10^4 N/m^2). The predicted rate is $\dot{\varepsilon}_{predict} = 1.1 \times 10^{-7}$ (min)$^{-1}$ (1.85×10^{-9} s^{-1}) compared with an observed rate $\dot{\varepsilon}_{observed} \sim 5 \times 10^{-7}$ (min)$^{-1}$ (8.3×10^{-9} s^{-1}). The close correspondence serves to demonstrate that the simple model is capable of predicting the right order of magnitude of the creep rate.

If dislocation motion controls Newtonian flow in pure Al then the addition of a finely dispersed phase in the Al matrix should have a large effect on the observed creep rate. To investigate this possibility several creep tests were performed on samples containing 0.5% Fe. The $FeAl_3$ precipitates in this alloy have a spacing, l, of about 10^{-3} cm. The stress required to bow a dislocation between two of these precipitates is then about 450 p.s.i. (3.1×10^6 N/m^2) assuming $\sigma = Gb/l$. Consequently any dislocation motion produced by applied stresses in

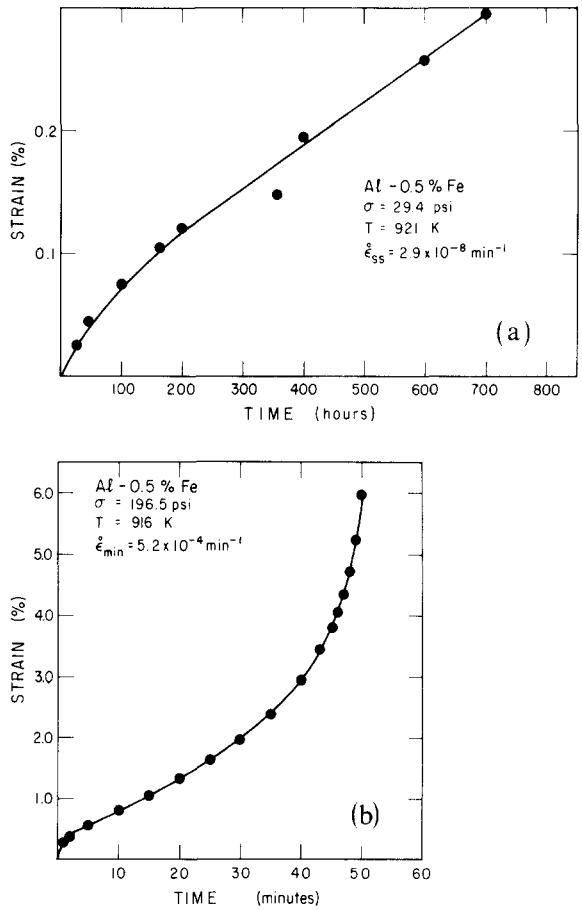

Fig. 8(a) Typical creep curve for Al–0.5% Fe tested at a stress less than 100 p.s.i. (6.89×10^5 N/m^2). (b) Typical creep curve for Al–0.5% Fe tested at a stress greater than 100 p.s.i. (6.89×10^5 N/m^2).

TABLE 5

Low stress–high temperature creep data for Al–0.5% Fe

Applied stress (*p.s.i.*)	(N/m^2)	*Temperature* (*K*)	*Creep rate* (min^{-1})	(s^{-1})
12	8.28×10^4	919	1.1×10^{-8}	1.83×10^{-10}
18	1.24×10^5	921	1.3×10^{-8}	2.17×10^{-10}
29	2.00×10^5	921	2.9×10^{-8}	4.83×10^{-10}
42	2.90×10^5	915	4.2×10^{-8}	7.00×10^{-10}
59	4.07×10^5	916	7.7×10^{-8}	1.28×10^{-9}
63	4.34×10^5	920	5.2×10^{-8}	8.67×10^{-10}
74	5.10×10^5	915	1.2×10^{-7}	2.00×10^{-9}
100	6.89×10^5	920	1.4×10^{-7}	2.33×10^{-9}
120	8.28×10^5	912	3.5×10^{-7}*	5.83×10^{-9}
175	1.21×10^6	915	2.1×10^{-5}*	3.50×10^{-7}
197	1.36×10^6	916	5.2×10^{-5}*	8.67×10^{-7}

* Steady state creep not observed. The minimum creep rate is reported.

the range of 1–100 p.s.i. (6.9×10^3–6.9×10^5 N/m^2) should be retarded by the precipitates.

Typical creep curves for the Al–0.5% Fe alloy are shown in Fig. 8. All samples tested at stresses lower than 100 p.s.i. showed a small amount of primary creep strain followed by an extensive steady state region. At stresses above 100 p.s.i. primary creep was followed by an extended tertiary region and in the following data only a minimum creep rate is reported. The observed creep rates as a function of applied stress are listed in Table 5 and plotted in Fig. 9 along with the data from pure Al. Below 100 p.s.i. (6.89×10^5 N/m^2) the Al–Fe alloy exhibits Newtonian viscosity with a creep rate ~ 100 times slower than that for pure Al. The stress dependence of the creep rate above 100 p.s.i. (6.89×10^5 N/m^2) probably has little physical significance as steady state was not observed.

Also shown on Fig. 9 is the theoretically predicted creep behavior of the Al–0.5% Fe assuming that directional diffusion is the rate-controlling mechanism. In contrast with the case for pure Al, the Al–0.5% Fe creep data fall below the predicted rate by a factor of 100. The predicted rates for Al–0.5% Fe are faster than the predicted rates for pure Al because of the small grain size (0.3 mm) for the alloy.

In addition to the poor absolute agreement between directional diffusion theory and experiment for Al–0.5% Fe, the temperature dependence of the creep rate is not equal to that for self diffusion. The data shown in Table 6, typical of several tests, indicate a creep activation energy several times that for self diffusion. This high creep activation energy, while not uncommonly observed for the creep of

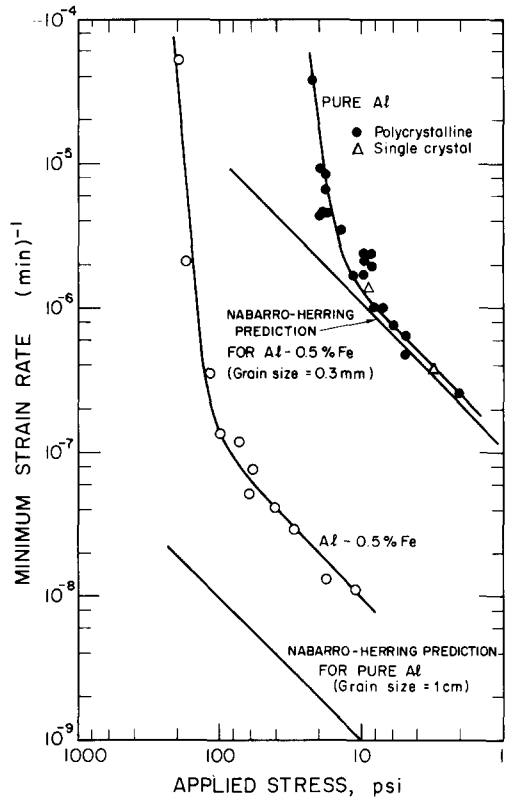

Fig. 9. Comparison of creep data for pure Al and Al–0.5% Fe with the predictions of the Nabarro–Herring theory. The theoretical predictions are made for a test temperature of 923K with the assumption that the temperature dependence for both materials is the same as that for Al self diffusion.

TABLE 6

Determination of creep activation energy for Al–0.5% Fe

Applied stress (*p.s.i.*)	(N/m^2)	*Temperature* *K*	*Creep rate* (min^{-1})	(s^{-1})	*Activation energy* (*kcal/mol*)
74	5.1×10^5	915	1.2×10^{-7}	2.00×10^{-9}	135
74	5.1×10^5	896	2.5×10^{-8}	4.16×10^{-10}	139
74	5.1×10^5	915	1.2×10^{-7}	2.00×10^{-9}	184
74	5.1×10^5	893	9.7×10^{-9}	1.62×10^{-10}	156
74	5.1×10^5	913	6.9×10^{-8}	1.15×10^{-9}	

two-phase materials, makes correlation of the data with theory complex because there is no simple thermally activated process requiring an activation energy of this magnitude.

Certainly the decrease of the Al–0.5% Fe creep rate, below that of pure Al, indicates that dislocation motion in the alloy has been impeded. Yet, it is not possible to state whether the deformation process in Al–0.5% Fe is controlled by dislocation motion as in the case of pure Al, or simply directional diffusion.

There have been other studies for which the calculated directional diffusion rates are faster than the observed creep rates. Harris *et al.*[27] and Burton[28] have observed this effect in metals with fine dispersions of a second phase. This behavior has been interpreted in terms of the second phase particles affecting the ease with which vacancies are generated or annihilated at grain boundaries[27,29]. Although the stress and temperature dependence of the strain rate predicted by these models have not been fully worked out, they do require a decrease in the creep rate from the Nabarro–Herring prediction.

SUMMARY OF CONCLUSIONS

The high temperature–low stress creep behavior of pure Al and Al–0.5% Fe has been examined. For Al it has been shown that:

1. Steady state strain rates are 1500 times greater than the Nabarro–Herring predicted rates
2. Single crystals creep at the same rate as polycrystals for the same stress and temperature
3. Pre-straining results in inverse transient behavior similar to that observed in cold worked metals tested at higher stresses and lower temperatures
4. The creep activation energy is nearly equal to the self diffusion activation energy
5. Subgrains form during creep
6. The dislocation density within subgrains is essentially stress independent.

A simple theory based on the climb controlled generation of dislocations from a fixed density of sources is developed to explain the observed behavior. In the case of Al–0.5% Fe we have shown that:

1. Steady state strain rates in the Newtonian range are 100 times slower than in pure Al at the same stress and temperature, in spite of the fact that the grain size in the Al–0.5% Fe alloy is smaller by a factor of 30
2. Steady state creep rates in the Newtonian range are some 100 times slower than the Nabarro–Herring theory
3. The creep activation energy greatly exceeds the diffusion activation energy.

The rate controlling mechanism for Newtonian flow in Al–0.5% Fe is unknown although the decrease in creep rate compared with pure Al supports the postulate that dislocation motion controls Newtonian flow in the pure material.

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Fluage à haute température et sous-faible contrainte de l'aluminium et de l'alliage Al–0,5% Fe

Les auteurs ont étudié la cinétique du fluage newtonien, à haute température et sous faible contrainte, de l'aluminium et de l'alliage Al–0,5% Fe. Ils ont examiné l'influence de la structure et comparé les résultats aux prédictions de la théorie de Nabarro–Herring (N–H) du fluage par diffusion directionnelle. Ils montrent que, bien que l'influence de la température et de la contrainte sur la vitesse de fluage newtonien de l'aluminium pur soit conforme à la théorie N–H, les vitesses absolues de fluage observées aussi bien dans les monocristaux que les polycristaux sont 1500 fois plus grandes que ne le prévoit la théorie. Ils montrent de plus que le fluage newtonien de l'aluminium est influencé par la structure. Au cours du fluage, la densité de dislocations augmente et il se forme des sous-grains. La dimension des sous grains diminue lorsque la contrainte augmente et les valeurs obtenues se raccordent parfaitement aux mesures de taille des sous-grains faites dans le cas de l'aluminium à température plus faible et pour des contraintes plus élevées. Par contre, la densité des dislocations est pratiquement indépendante de la contrainte. Une prédéformation à basse température provoque un fluage inverse transitoire, analogue à celui que l'on observe dans les métaux écrouis à froid et déformés à température plus faible et sous contrainte plus élevée que dans la présente étude. D'autre part, la présence de précipités de $FeAl_3$ diminue la vitesse de fluage de l'aluminium dans un rapport de 1 à 100, pour une contrainte et une température identiques. Cependant, dans l'alliage Al–0,5% Fe, les grains sont 30 fois plus petits que dans l'aluminium pur. Les faits expérimentaux conduisent les auteurs à conclure que le fluage newtonien de l'aluminium est régi par un mécanisme de dislocations plutôt que par la diffusion directionnelle. Pour expliquer le comportement observé, ils exposent une théorie simple, fondée sur la création de dislocations à partir d'un nombre constant de sources selon un mécanisme contrôlé par les phénomènes de montée.

Kriechen von Al und Al–0,5% Fe bei hohen Temperaturen und kleinen Spannungen

Die Kinetik und Strukturabhängigkeit des Newtonschen Kriechens von Al und Al + 0,5% Fe bei hohen Temperaturen und kleinen Spannungen wurde untersucht und mit den Vorhersagen der Nabarro–Herring (N–H)-Theorie der gerichteten Diffusion verglichen. Es wird gezeigt, daß zwar die Spannungs- und Temperaturabhängigkeit der Newtonschen Kriechgeschwindigkeit von reinem Al in Übereinstimmung mit der N–H-Theorie sind, daß jedoch die beobachteten Kriechgeschwindigkeiten sowohl von Einkristallen als auch von Vielkristallen etwa 1500 mal größer sind als die Theorie vorhersagt. Außerdem wird gezeigt, daß Newtonsches Fließen in Al strukturempfindlich ist. Während des Kriechens nimmt die Versetzungsdichte zu und es bilden sich Subkörner. Die Größe der Subkörner ist umgekehrt proportional zur Spannung; eine Extrapolation der Subkorngrößen auf kleine Temperaturen und höhere Spannungen liefert vernünftige Werte. Die Versetzungsdichte ist im wesentlichen spannungsunabhängig. Vorverformung bei tiefen Temperaturen führt zu einem inversen Verhalten beim Übergangskriechen, ähnlich wie es an kaltverformten Metallen in Kriechversuchen bei tieferen Temperaturen und höheren Spannungen beobachtet wurde. Bei gleicher Temperatur und Spannung wird die Kriechgeschwindigkeit von Al durch die Anwesenheit von $FeAl_3$-Ausscheidungen etwa um einen Faktor 100 verkleinert, obwohl die Korngröße in der Al–0,5% Fe-Legierung um einen Faktor 30 kleiner ist. Diese experimentellen Beobachtungen führen uns zu der Schlußfolgerung, daß Newtonsches Kriechen von Al durch einen Versetzungsmechanismus und nicht durch gerichtete Diffusion kontrolliert wird. Zur Erklärung des beobachteten Verhaltens wird eine einfache Theorie entwickelt; als Grundlage dient die durch einen Kletterprozeß kontrollierte Erzeugung neuer Versetzungen aus einer unveränderten Dichte von Versetzungsquellen.